Course Project 2026

Unison: A Learning Management System with Collaborative Group Submissions

(Milestone 2: ER Diagram and Data Model)

Course Code : CS602C
Course Name : Advanced Software Development Lab
Course Instructor : Prof. Gayathri Ananthanarayanan
Team : Team Bootstrap
Repository : https://github.com/sandeepmohaniitdh/unison

Team Details

Name	Roll Number	Email
Sandeep M	CS26MT001	cs26mt001@iitdh.ac.in
Km Shristi Singh	CS26MT011	cs26mt011@iitdh.ac.in
Tirtho Paul	CS26MT024	cs26mt024@iitdh.ac.in
Ujjwal Gupta	CS26MT025	cs26mt025@iitdh.ac.in

॥ सा विद्या या विमुक्तये ॥
ಭಾ.ತಂ.ಸಂ. ಧಾರವಾಡ
भा. प्रौ. सं. धारवाड
I.I.T. DHARWAD

Department of Computer Science and Engineering

Indian Institute of Technology, Dharwad

Contents

1 About this data model **2**
1.1 Conventions . . . 2
1.2 Notation . . . 2

2 Overview diagram **4**

3 Detailed diagrams **5**
3.1 Identity and sessions . . . 5
3.2 Departments, courses and enrolment . . . 6
3.3 Assignments, teams and submissions . . . 7
3.4 Review, marks and AI feedback . . . 8
3.5 Notifications, email and administration . . . 9

4 Relationships and cardinalities **10**

5 Normalisation **11**

6 Expected indexes **12**

A Data dictionary **14**
A.1 Identity and sessions . . . 14
A.2 Departments, courses and enrolment . . . 16
A.3 Assignments, teams and submissions . . . 18
A.4 Review, marks and AI feedback . . . 20
A.5 Notifications and email . . . 22
A.6 Administration . . . 23

1 About this data model

This document gives the entity-relationship (ER) model behind Unison: 29 tables in six groups, matching the design document submitted at MS-1. It contains an overview diagram, one detailed diagram per group with every column, primary key, foreign key and cardinality, a list of all 48 relationships, the reasoning for third normal form, and the indexes we expect to need. Appendix A is a full data dictionary.

1.1 Conventions

- **Engine and text:** MySQL 8, InnoDB, `utf8mb4`. All tables are created by Alembic migrations.
- **Keys:** most tables have a surrogate primary key `id` (`BIGINT UNSIGNED AUTO_INCREMENT`). Link tables whose identity is a pair of references use that pair as a composite primary key.
- **Times:** every timestamp is `DATETIME(3)` in UTC. Deadlines are converted from and to IST (Asia/Kolkata) by the API.
- **Fixed sets** such as roles and statuses are `ENUM`s, because changing them needs code changes anyway. Notification event types are `VARCHAR`, because new ones are added often.
- **Files** are not stored in the database. Rows hold the original name, type, size, a random storage key and, for submissions, a SHA-256 hash.
- **Nothing a user did is hard-deleted.** Accounts are anonymised after the 7-day grace period, and courses are archived, so foreign keys to `users` and `courses` use `ON DELETE RESTRICT`. Rows that only exist inside their parent (emails, sessions, tokens, rubric lines, scores, adjustments, files, preferences) use `ON DELETE CASCADE`.
- **Computed, not stored:** team totals, each member's mark, team size, a submission's status and unread counts are calculated by queries (Section 5).

1.2 Notation

The diagrams use crow's-foot notation. On each line, the symbol next to a table says how many rows of that table relate to one row at the other end: the inner mark is the maximum and the outer mark the minimum. Key badges in the detailed diagrams are **PK** (primary key), **FK** (foreign key) and **UQ** (part of a unique key); `NULL` marks optional columns. Tables drawn with a dashed border belong to another group and are shown in full on that group's page.

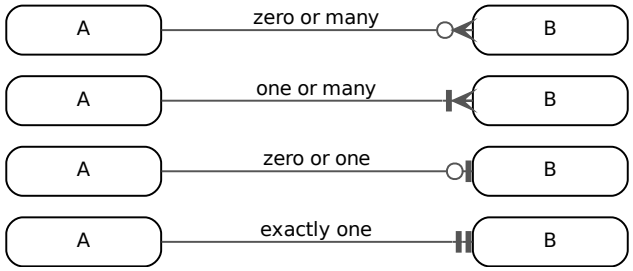


Figure 1: Crow's-foot symbols used in the diagrams.

Table 1: The six groups of tables.

Group	**Tables**
Identity and sessions	`users`, `user_emails`, `oauth_identities`, `sessions`, `login_attempts`, `email_tokens`
Departments, courses and enrolment	`departments`, `courses`, `enrollments`, `invitations`, `resources`
Assignments, teams and submissions	`assignments`, `rubric_criteria`, `teams`, `team_members`, `submissions`, `submission_versions`, `final_submissions`, `submission_files`
Review, marks and AI feedback	`reviews`, `review_scores`, `member_adjustments`, `ai_feedback`
Notifications and email	`notifications`, `notification_preferences`, `notification_rules`, `email_outbox`
Administration	`audit_log`, `platform_settings`

2 Overview diagram

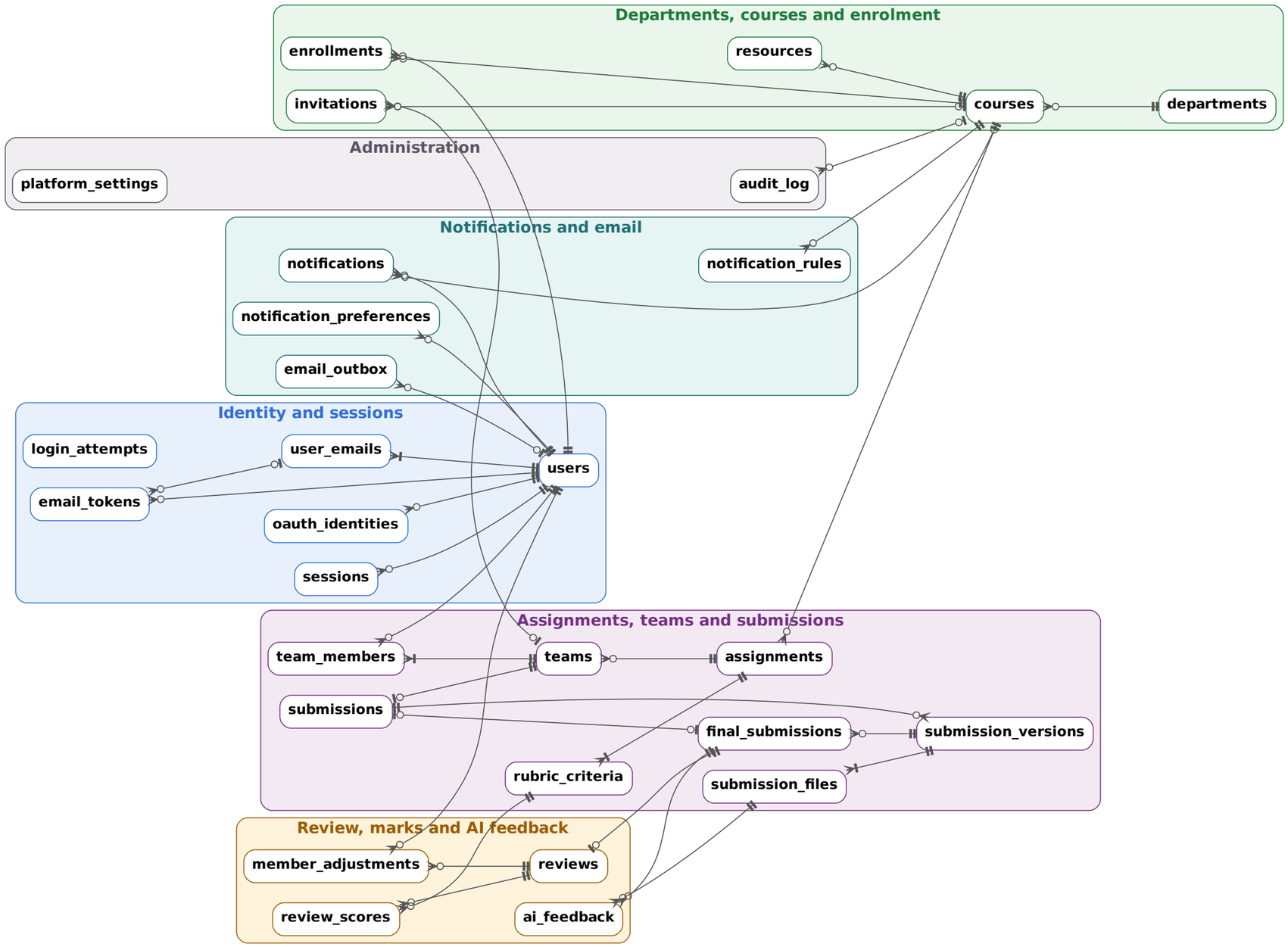


Figure 2: All 29 tables and their relationships. References that only record who did something (such as `created_by` or `submitted_by`) are left out here for readability and appear in the detailed diagrams.

3 Detailed diagrams

3.1 Identity and sessions

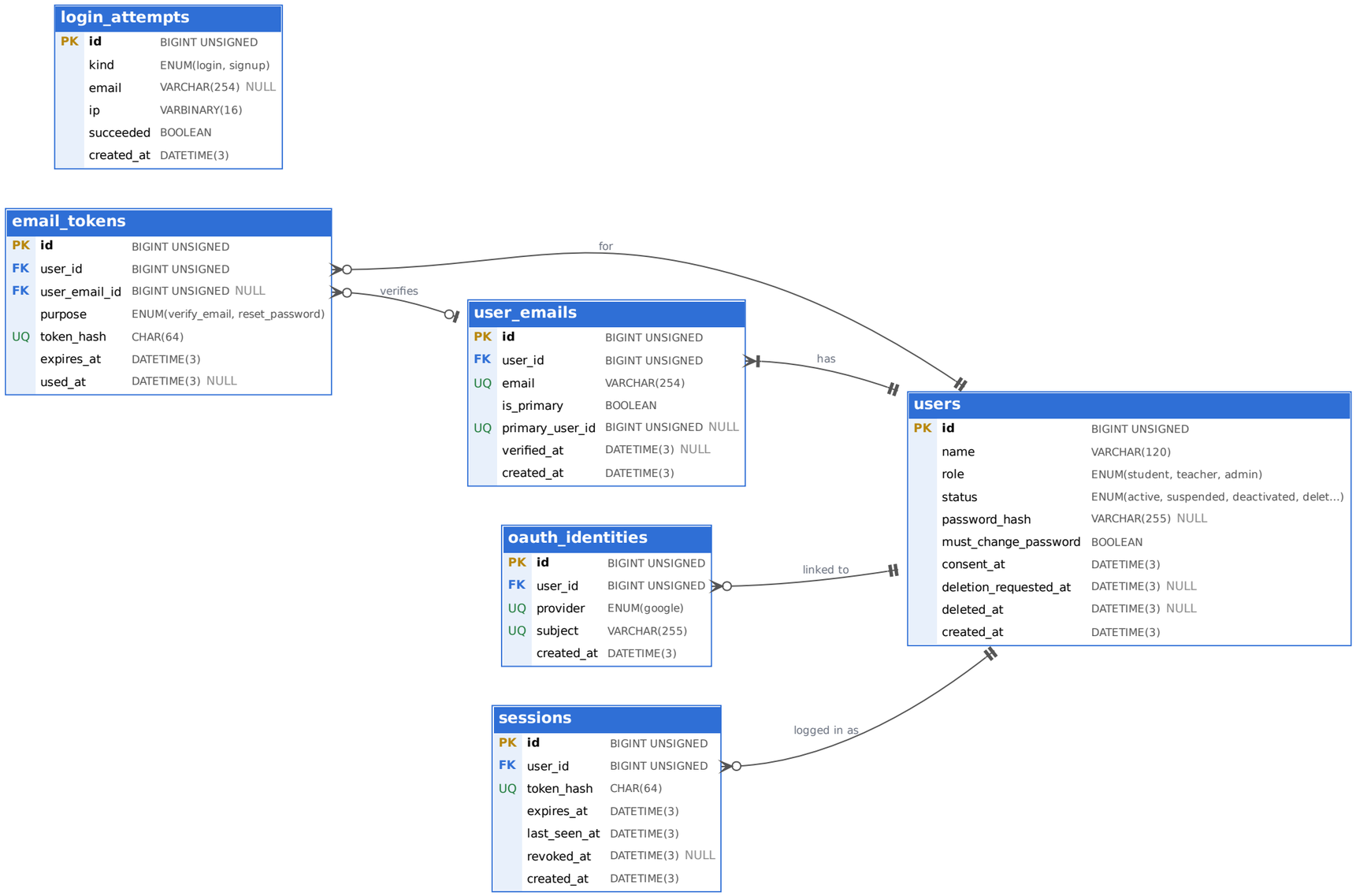


Figure 3: Identity and sessions. Lines start at the foreign-key column and end at the referenced table; dashed tables belong to another group.

3.2 Departments, courses and enrolment

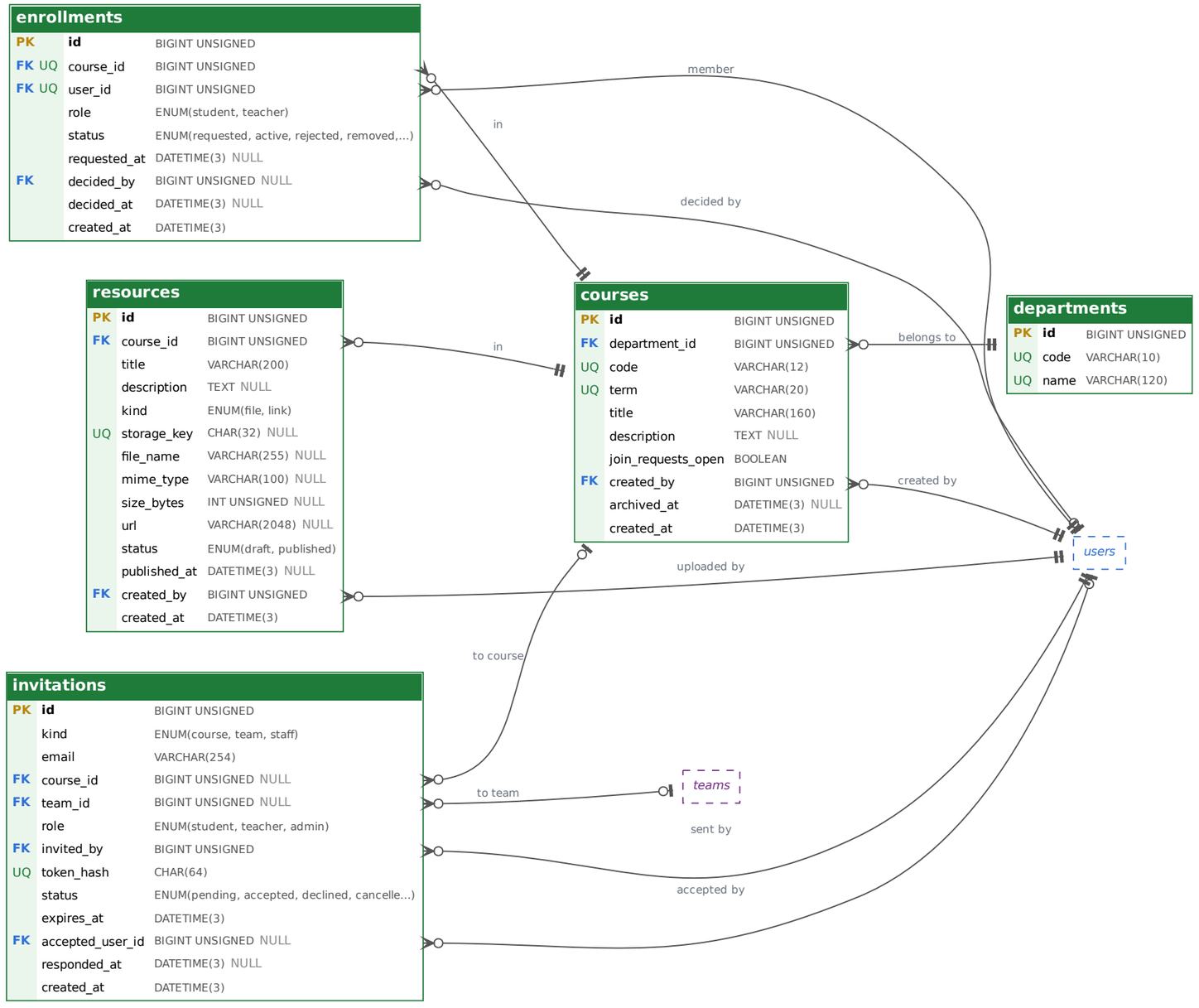


Figure 4: Departments, courses and enrolment. Lines start at the foreign-key column and end at the referenced table; dashed tables belong to another group.

3.3 Assignments, teams and submissions

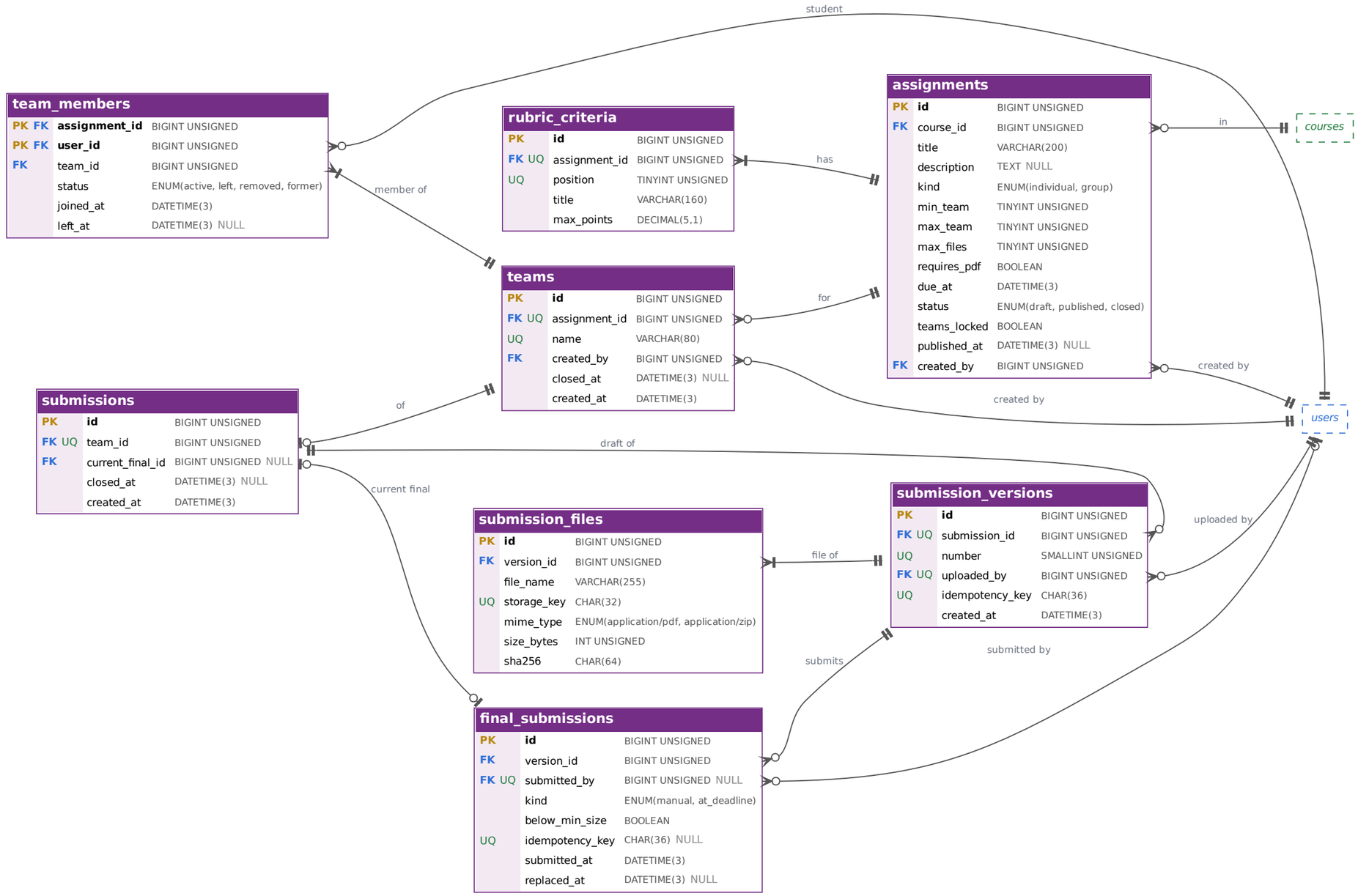


Figure 5: Assignments, teams and submissions. Lines start at the foreign-key column and end at the referenced table; dashed tables belong to another group.

3.4 Review, marks and AI feedback

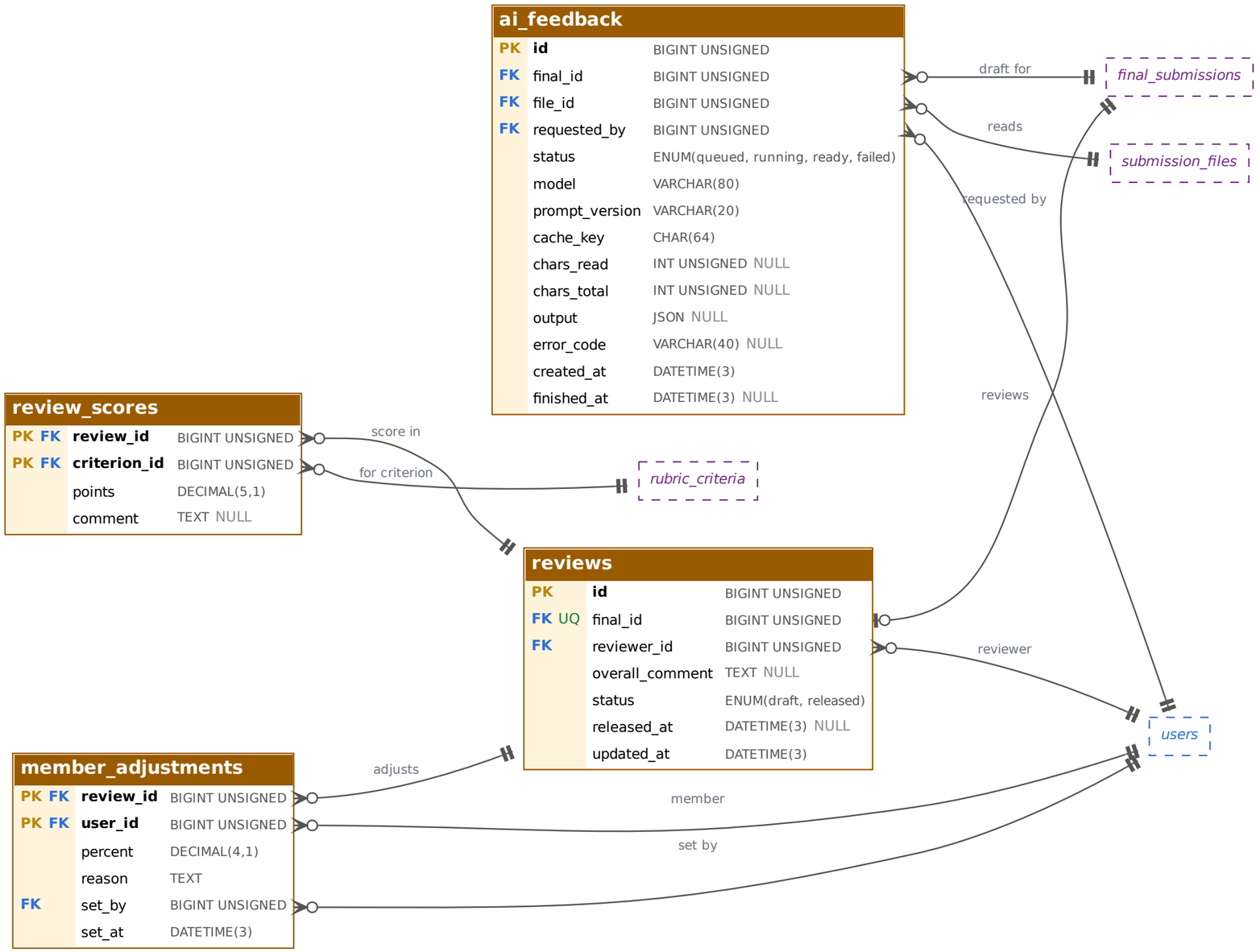


Figure 6: Review, marks and AI feedback. Lines start at the foreign-key column and end at the referenced table; dashed tables belong to another group.

3.5 Notifications, email and administration

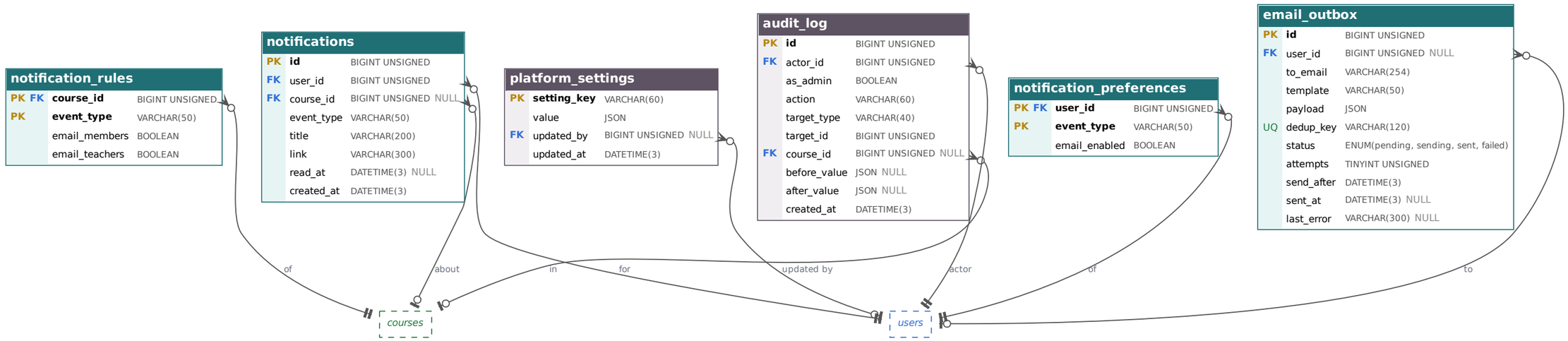


Figure 7: Notifications, email and administration. Lines start at the foreign-key column and end at the referenced table; dashed tables belong to another group.

4 Relationships and cardinalities

Every foreign key in the model. Read a row as: each referencing row points to *(column 3)* referenced rows, and each referenced row has *(column 4)* referencing rows. Multiplicities are written 1 (exactly one), 0..1 (zero or one), 0..* (zero or many) and 1..* (one or many).

Table 2: All 48 relationships.

Foreign key	References	Each row points to	Each target has	Meaning
user_emails.user_id	users.id	1	1..*	has
oauth_identities.user_id	users.id	1	0..*	linked to
sessions.user_id	users.id	1	0..*	logged in as
email_tokens.user_id	users.id	1	0..*	for
email_tokens.user_email_id	user_emails.id	0..1	0..*	verifies
courses.department_id	departments.id	1	0..*	belongs to
courses.created_by	users.id	1	0..*	created by
enrollments.course_id	courses.id	1	0..*	in
enrollments.user_id	users.id	1	0..*	member
enrollments.decided_by	users.id	0..1	0..*	decided by
invitations.course_id	courses.id	0..1	0..*	to course
invitations.team_id	teams.id	0..1	0..*	to team
invitations.invited_by	users.id	1	0..*	sent by
invitations.accepted_user_id	users.id	0..1	0..*	accepted by
resources.course_id	courses.id	1	0..*	in
resources.created_by	users.id	1	0..*	uploaded by
assignments.course_id	courses.id	1	0..*	in
assignments.created_by	users.id	1	0..*	created by
rubric_criteria.assignment_id	assignments.id	1	1..*	has
teams.assignment_id	assignments.id	1	0..*	for
teams.created_by	users.id	1	0..*	created by
team_members.team_id, assignment_id	teams.id, assignment_id	1	1..*	member of
team_members.user_id	users.id	1	0..*	student
submissions.team_id	teams.id	1	0..1	of
submissions.current_final_id	final_submissions.id	0..1	0..1	current final
submission_versions.submission_id	submissions.id	1	0..*	draft of
submission_versions.uploaded_by	users.id	1	0..*	uploaded by
final_submissions.version_id	submission_versions.id	1	0..*	submits
final_submissions.submitted_by	users.id	0..1	0..*	submitted by
submission_files.version_id	submission_versions.id	1	1..*	file of
reviews.final_id	final_submissions.id	1	0..1	reviews
reviews.reviewer_id	users.id	1	0..*	reviewer
review_scores.review_id	reviews.id	1	0..*	score in
review_scores.criterion_id	rubric_criteria.id	1	0..*	for criterion
member_adjustments.review_id	reviews.id	1	0..*	adjusts
member_adjustments.user_id	users.id	1	0..*	member
member_adjustments.set_by	users.id	1	0..*	set by
ai_feedback.final_id	final_submissions.id	1	0..*	draft for
ai_feedback.file_id	submission_files.id	1	0..*	reads
ai_feedback.requested_by	users.id	1	0..*	requested by
notifications.user_id	users.id	1	0..*	for

Table 2 continued

Foreign key	References	Each row points to	Each target has	Meaning
notifications.course_id	courses.id	0..1	0..*	about
notification_preferences.user_id	users.id	1	0..*	of
notification_rules.course_id	courses.id	1	0..*	of
email_outbox.user_id	users.id	0..1	0..*	to
audit_log.actor_id	users.id	1	0..*	actor
audit_log.course_id	courses.id	0..1	0..*	in
platform_settings.updated_by	users.id	0..1	0..*	updated by

Three relationships carry the main business rules:

- **One team per student per assignment.** team_members has the primary key (assignment_id, user_id), and the composite foreign key (team_id, assignment_id) references teams(id, assignment_id). The database itself refuses a second team for the same student, and the assignment in the row always matches the team's.
- **The teacher sees only the current final.** submissions.current_final_id points to one row of final_submissions; replaced finals stay in that table with replaced_at set, visible only to the team.
- **One primary email per user.** The generated column user_emails.primary_user_id equals user_id for the primary email and is NULL for every other email. A unique index on it allows any number of NULLs, so each user has exactly one primary email and as many verified second emails as needed.

5 Normalisation

All 29 tables are in third normal form (3NF): every column holds a single value, every non-key column depends on the whole key, and no non-key column depends on another non-key column. The decisions that keep it that way:

- **No derived values are stored.** A team's mark is the sum of its review_scores; a member's mark is that sum plus their member_adjustments.percent, kept between 0 and the maximum; team size is a count of active team_members; a submission's status (no work, drafts, final, submitted at deadline, not submitted) follows from its versions, current_final_id and closed_at. Storing any of these would create a dependency on other rows that could go stale.
- final_submissions **has no** submission_id**.** The submission is reached through version_id; storing it as well would make it depend on version_id, a transitive dependency.
- reviews **references the final, not the team or assignment,** for the same reason: both follow from final_id.
- team_members **is 3NF but not BCNF, on purpose.** Its candidate keys are (assignment_id, user_id) and (team_id, user_id). The dependency team_id → assignment_id has a prime attribute on its right side, which 3NF allows. Keeping assignment_id in the key is what lets MySQL enforce one team per student per assignment, and the composite foreign key means it can never disagree with the team.

- **Emails are a separate table.** A user can have a primary and a second email, each with its own verification time, so emails are not columns of `users`.
- **Teachers of a course are enrolment rows** with `role = teacher`, not a column of `courses`, because a course can have several teachers.
- **Polymorphic invitations stay atomic.** An invitation has either a `course_id` or a `team_id` depending on `kind`, enforced by a `CHECK` constraint; both are plain foreign keys, not a combined "target" string.
- `audit_log` **keeps before and after values as JSON** because it is an append-only history of many different tables; nothing is ever read back from it into another table.

6 Expected indexes

Primary keys and unique keys are indexed automatically, and InnoDB indexes every foreign key. Table 3 lists the indexes the main queries need beyond that, or the unique keys that carry a business rule.

Table 3: Expected indexes and the queries they serve.

Table	Index	Serves
`user_emails`	`UNIQUE (email)`	find an account by any email at login and when accepting invitations
`user_emails`	`UNIQUE (primary_user_id)`	exactly one primary email per user; NULL for second emails
`sessions`	`UNIQUE (token_hash)`	look up the session on every request
`login_attempts`	`(email, ip, created_at), (ip, created_at)`	count recent failures per email-and-IP pair and per IP
`courses`	`UNIQUE (code, term); FULLTEXT (code, title)`	catalogue search by code or title
`courses`	`(department_id)`	filter the catalogue by department
`enrollments`	`UNIQUE (course_id, user_id); (user_id, status)`	one row per person per course; a user's courses on the dashboard
`enrollments`	`(course_id, status)`	pending join requests and the member list
`invitations`	`UNIQUE (token_hash); (team_id, status); (email, status)`	open a link; count a team's pending invitations; a user's pending invitations
`resources`	`(course_id, status, published_at)`	the Resources section in order
`assignments`	`(course_id, status, due_at); (status, due_at)`	a course's assignments; the worker's deadline scan
`teams`	`UNIQUE (assignment_id, name); UNIQUE (id, assignment_id)`	team names per assignment; target of the composite FK
`team_members`	`PRIMARY KEY (assignment_id, user_id); (team_id, status)`	one team per student per assignment; list a team's members
`submission_versions`	`UNIQUE (submission_id, number); UNIQUE (uploaded_by, idempotency_key)`	draft numbering; duplicate-click protection
`final_submissions`	`(version_id); UNIQUE (submitted_by, idempotency_key)`	history of finals; duplicate-click protection
`submission_files`	`UNIQUE (storage_key); (version_id)`	downloads; files of a draft
`reviews`	`UNIQUE (final_id)`	one review per final

Table 3 continued

Table	Index	Serves
ai_feedback	(cache_key, status); (final_id, created_at)	reuse a cached draft; count requests per submission
notifications	(user_id, read_at, created_at)	unread count for the bell and the dashboard list
email_outbox	UNIQUE (dedup_key); (status, send_after)	no duplicate emails; the worker's next batch
audit_log	(created_at); (actor_id, created_at); (target_type, target_id)	filter the audit log by time, actor or record

Other integrity rules

- CHECK constraints: assignments.max_team >= min_team; rubric_criteria.max_points > 0; member_adjustments.percent BETWEEN -20 AND 10; and an invitation's course_id or team_id must match its kind.
- Rules that need other rows are checked in the service layer inside the same transaction, under a row lock where races are possible: a score must not exceed its criterion's max_points; members plus pending invitations must not exceed max_team; a final needs the minimum team size before the deadline.

A Data dictionary

Every column of every table. "Key" lists PK, FK and UQ (part of a unique key); "Null" is "yes" where the column is optional.

A.1 Identity and sessions

`users`

Every account; the email lives in user_emails.

Column	Type	Key	Null	Note
id	BIGINT UNSIGNED	PK		
name	VARCHAR(120)			replaced by 'Former member' after deletion
role	ENUM(student, teacher, admin)			global role
status	ENUM(active, suspended, deactivated, deleted)			
password_hash	VARCHAR(255)		yes	Argon2; NULL for Google-only accounts
must_change_password	BOOLEAN			true for the first admin
consent_at	DATETIME(3)			data-consent notice accepted
deletion_requested_at	DATETIME(3)		yes	starts the 7-day grace period
deleted_at	DATETIME(3)		yes	set when anonymised
created_at	DATETIME(3)			

`user_emails`

Primary and second emails; invitations match any verified one.

Column	Type	Key	Null	Note
id	BIGINT UNSIGNED	PK		
user_id	BIGINT UNSIGNED	FK		
email	VARCHAR(254)	UQ		one owner per address
is_primary	BOOLEAN			
primary_user_id	BIGINT UNSIGNED	UQ	yes	generated: user_id if is_primary, else NULL
verified_at	DATETIME(3)		yes	
created_at	DATETIME(3)			

`oauth_identities`

Linked Google accounts.

Column	Type	Key	Null	Note
id	BIGINT UNSIGNED	PK		
user_id	BIGINT UNSIGNED	FK		
provider	ENUM(google)	UQ		
subject	VARCHAR(255)	UQ		Google 'sub' claim
created_at	DATETIME(3)			

`sessions`

Server-side sessions; only the token hash is stored.

Column	Type	Key	Null	Note
id	BIGINT UNSIGNED	PK		
user_id	BIGINT UNSIGNED	FK		
token_hash	CHAR(64)	UQ		SHA-256 of the cookie value
expires_at	DATETIME(3)			7 days
last_seen_at	DATETIME(3)			
revoked_at	DATETIME(3)		yes	logout, deletion
created_at	DATETIME(3)			

`login_attempts`

Login and sign-up attempts for rate limits; no FK on purpose.

Column	Type	Key	Null	Note
id	BIGINT UNSIGNED	PK		
kind	ENUM(login, signup)			
email	VARCHAR(254)		yes	as typed; may not exist
ip	VARBINARY(16)			IPv4 or IPv6
succeeded	BOOLEAN			
created_at	DATETIME(3)			

`email_tokens`

Single-use links for email verification and password reset.

Column	Type	Key	Null	Note
id	BIGINT UNSIGNED	PK		
user_id	BIGINT UNSIGNED	FK		
user_email_id	BIGINT UNSIGNED	FK	yes	the address being verified
purpose	ENUM(verify_email, reset_password)			
token_hash	CHAR(64)	UQ		
expires_at	DATETIME(3)			30 min for reset
used_at	DATETIME(3)		yes	

A.2 Departments, courses and enrolment

`departments`

Departments managed by the administrator.

Column	Type	Key	Null	Note
id	BIGINT UNSIGNED	PK		
code	VARCHAR(10)	UQ		e.g. CSE
name	VARCHAR(120)	UQ		

`courses`

A course offering in one term.

Column	Type	Key	Null	Note
id	BIGINT UNSIGNED	PK		
department_id	BIGINT UNSIGNED	FK		
code	VARCHAR(12)	UQ		e.g. CS301
term	VARCHAR(20)	UQ		e.g. Jul-Nov 2026
title	VARCHAR(160)			
description	TEXT		yes	
join_requests_open	BOOLEAN			
created_by	BIGINT UNSIGNED	FK		teacher or admin
archived_at	DATETIME(3)		yes	read-only once set
created_at	DATETIME(3)			

`enrollments`

Course membership, teachers included; join requests are rows in 'requested'.

Column	Type	Key	Null	Note
id	BIGINT UNSIGNED	PK		
course_id	BIGINT UNSIGNED	FK, UQ		
user_id	BIGINT UNSIGNED	FK, UQ		
role	ENUM(student, teacher)			role in this course
status	ENUM(requested, active, rejected, removed, left)			
requested_at	DATETIME(3)		yes	
decided_by	BIGINT UNSIGNED	FK	yes	teacher or admin who approved, rejected or removed
decided_at	DATETIME(3)		yes	
created_at	DATETIME(3)			

`invitations`

Course, team and staff invitations sent by email.

Column	Type	Key	Null	Note
id	BIGINT UNSIGNED	PK		
kind	ENUM(course, team, staff)			
email	VARCHAR(254)			invited address
course_id	BIGINT UNSIGNED	FK	yes	kind = course
team_id	BIGINT UNSIGNED	FK	yes	kind = team
role	ENUM(student, teacher, admin)			role granted on accept
invited_by	BIGINT UNSIGNED	FK		
token_hash	CHAR(64)	UQ		
status	ENUM(pending, accepted, declined, cancelled, expired)			
expires_at	DATETIME(3)			7 days
accepted_user_id	BIGINT UNSIGNED	FK	yes	
responded_at	DATETIME(3)		yes	
created_at	DATETIME(3)			

`resources`

Notes, slides and links; drafts until published.

Column	Type	Key	Null	Note
id	BIGINT UNSIGNED	PK		
course_id	BIGINT UNSIGNED	FK		
title	VARCHAR(200)			
description	TEXT		yes	
kind	ENUM(file, link)			
storage_key	CHAR(32)	UQ	yes	random file name; kind = file
file_name	VARCHAR(255)		yes	original name
mime_type	VARCHAR(100)		yes	from the allowlist
size_bytes	INT UNSIGNED		yes	max 25 MB
url	VARCHAR(2048)		yes	kind = link
status	ENUM(draft, published)			
published_at	DATETIME(3)		yes	
created_by	BIGINT UNSIGNED	FK		
created_at	DATETIME(3)			

A.3 Assignments, teams and submissions

`assignments`

Individual or group assignments with a deadline in UTC (shown in IST).

Column	Type	Key	Null	Note
id	BIGINT UNSIGNED	PK		
course_id	BIGINT UNSIGNED	FK		
title	VARCHAR(200)			
description	TEXT		yes	
kind	ENUM(individual, group)			individual = team of one
min_team	TINYINT UNSIGNED			1 for individual
max_team	TINYINT UNSIGNED			CHECK max_team ≥ min_team
max_files	TINYINT UNSIGNED			default 5
requires_pdf	BOOLEAN			a final must contain a PDF
due_at	DATETIME(3)			UTC
status	ENUM(draft, published, closed)			
teams_locked	BOOLEAN			
published_at	DATETIME(3)		yes	
created_by	BIGINT UNSIGNED	FK		

`rubric_criteria`

Rubric lines of an assignment.

Column	Type	Key	Null	Note
id	BIGINT UNSIGNED	PK		
assignment_id	BIGINT UNSIGNED	FK, UQ		
position	TINYINT UNSIGNED	UQ		display order
title	VARCHAR(160)			
max_points	DECIMAL(5,1)			CHECK > 0

`teams`

A team for one assignment.

Column	Type	Key	Null	Note
id	BIGINT UNSIGNED	PK, UQ		
assignment_id	BIGINT UNSIGNED	FK, UQ		UNIQUE(id, assignment_id) is the target of a composite FK
name	VARCHAR(80)	UQ		
created_by	BIGINT UNSIGNED	FK		recorded only; no extra rights
closed_at	DATETIME(3)		yes	last member left before a final
created_at	DATETIME(3)			

`team_members`

A student's place in an assignment: one row per student per assignment.

Column	Type	Key	Null	Note
`assignment_id`	`BIGINT UNSIGNED`	PK, FK		PK(assignment_id, user_id): one team per student per assignment
`user_id`	`BIGINT UNSIGNED`	PK, FK		
`team_id`	`BIGINT UNSIGNED`	FK		composite FK (team_id, assignment_id) to teams
`status`	`ENUM(active, left, removed, former)`			former = account deleted
`joined_at`	`DATETIME(3)`			
`left_at`	`DATETIME(3)`		yes	

`submissions`

The single submission record of a team.

Column	Type	Key	Null	Note
`id`	`BIGINT UNSIGNED`	PK		
`team_id`	`BIGINT UNSIGNED`	FK, UQ		one per team
`current_final_id`	`BIGINT UNSIGNED`	FK	yes	the final the teacher sees
`closed_at`	`DATETIME(3)`		yes	set when the deadline is applied
`created_at`	`DATETIME(3)`			

`submission_versions`

Team drafts, numbered per submission.

Column	Type	Key	Null	Note
`id`	`BIGINT UNSIGNED`	PK		
`submission_id`	`BIGINT UNSIGNED`	FK, UQ		
`number`	`SMALLINT UNSIGNED`	UQ		v1, v2, ...
`uploaded_by`	`BIGINT UNSIGNED`	FK, UQ		
`idempotency_key`	`CHAR(36)`	UQ		a repeated click returns the first result
`created_at`	`DATETIME(3)`			

`final_submissions`

Every time a draft became the final, including replaced finals.

Column	Type	Key	Null	Note
`id`	`BIGINT UNSIGNED`	PK		
`version_id`	`BIGINT UNSIGNED`	FK		the draft submitted
`submitted_by`	`BIGINT UNSIGNED`	FK, UQ	yes	NULL when submitted at the deadline
`kind`	`ENUM(manual, at_deadline)`			
`below_min_size`	`BOOLEAN`			flag for the teacher
`idempotency_key`	`CHAR(36)`	UQ	yes	
`submitted_at`	`DATETIME(3)`			
`replaced_at`	`DATETIME(3)`		yes	

`submission_files`

Files of a draft (PDF or ZIP, never extracted).

Column	Type	Key	Null	Note
`id`	`BIGINT UNSIGNED`	PK		
`version_id`	`BIGINT UNSIGNED`	FK		
`file_name`	`VARCHAR(255)`			original name
`storage_key`	`CHAR(32)`	UQ		random name on the volume
`mime_type`	`ENUM(application/pdf, application/zip)`			
`size_bytes`	`INT UNSIGNED`			max 10 MB
`sha256`	`CHAR(64)`			AI cache key input

A.4 Review, marks and AI feedback

`reviews`

The teacher's review of one final; totals are computed, not stored.

Column	Type	Key	Null	Note
`id`	`BIGINT UNSIGNED`	PK		
`final_id`	`BIGINT UNSIGNED`	FK, UQ		one review per final
`reviewer_id`	`BIGINT UNSIGNED`	FK		teacher or admin
`overall_comment`	`TEXT`		yes	
`status`	`ENUM(draft, released)`			
`released_at`	`DATETIME(3)`		yes	
`updated_at`	`DATETIME(3)`			

`review_scores`

Points and comment per rubric criterion.

Column	Type	Key	Null	Note
review_id	BIGINT UNSIGNED	PK, FK		
criterion_id	BIGINT UNSIGNED	PK, FK		
points	DECIMAL(5,1)			0 to max_points
comment	TEXT		yes	

`member_adjustments`

Optional per-member adjustment of the team mark.

Column	Type	Key	Null	Note
review_id	BIGINT UNSIGNED	PK, FK		
user_id	BIGINT UNSIGNED	PK, FK		the member
percent	DECIMAL(4,1)			CHECK BETWEEN -20 AND 10
reason	TEXT			shown to the member
set_by	BIGINT UNSIGNED	FK		
set_at	DATETIME(3)			

`ai_feedback`

AI feedback drafts; also the job queue for AI work.

Column	Type	Key	Null	Note
id	BIGINT UNSIGNED	PK		
final_id	BIGINT UNSIGNED	FK		
file_id	BIGINT UNSIGNED	FK		the PDF that was read
requested_by	BIGINT UNSIGNED	FK		
status	ENUM(queued, running, ready, failed)			
model	VARCHAR(80)			
prompt_version	VARCHAR(20)			
cache_key	CHAR(64)			hash of PDF, rubric, prompt, model
chars_read	INT UNSIGNED		yes	for the truncation warning
chars_total	INT UNSIGNED		yes	
output	JSON		yes	suggested points and comments
error_code	VARCHAR(40)		yes	
created_at	DATETIME(3)			
finished_at	DATETIME(3)		yes	

A.5 Notifications and email

`notifications`

In-app notifications (always on).

Column	Type	Key	Null	Note
id	BIGINT UNSIGNED	PK		
user_id	BIGINT UNSIGNED	FK		recipient
course_id	BIGINT UNSIGNED	FK	yes	
event_type	VARCHAR(50)			e.g. final_submitted
title	VARCHAR(200)			
link	VARCHAR(300)			page to open
read_at	DATETIME(3)		yes	
created_at	DATETIME(3)			

`notification_preferences`

A user's email switch per event type.

Column	Type	Key	Null	Note
user_id	BIGINT UNSIGNED	PK, FK		
event_type	VARCHAR(50)	PK		
email_enabled	BOOLEAN			default true

`notification_rules`

A course's email rule per event type.

Column	Type	Key	Null	Note
course_id	BIGINT UNSIGNED	PK, FK		
event_type	VARCHAR(50)	PK		
email_members	BOOLEAN			
email_teachers	BOOLEAN			

`email_outbox`

Transactional outbox; the worker sends and retries.

Column	Type	Key	Null	Note
id	BIGINT UNSIGNED	PK		
user_id	BIGINT UNSIGNED	FK	yes	NULL for invitees without an account
to_email	VARCHAR(254)			
template	VARCHAR(50)			
payload	JSON			
dedup_key	VARCHAR(120)	UQ		same email never queued twice
status	ENUM(pending, sending, sent, failed)			
attempts	TINYINT UNSIGNED			retries at 1, 5, 15, 60 min
send_after	DATETIME(3)			10-minute bundling for resources
sent_at	DATETIME(3)		yes	
last_error	VARCHAR(300)		yes	

A.6 Administration

`audit_log`

Every admin action and every role change; append-only.

Column	Type	Key	Null	Note
id	BIGINT UNSIGNED	PK		
actor_id	BIGINT UNSIGNED	FK		
as_admin	BOOLEAN			acting with admin powers in a course
action	VARCHAR(60)			e.g. role_changed
target_type	VARCHAR(40)			table name
target_id	BIGINT UNSIGNED			
course_id	BIGINT UNSIGNED	FK	yes	
before_value	JSON		yes	
after_value	JSON		yes	
created_at	DATETIME(3)			

`platform_settings`

Key-value settings, e.g. allowed email domain, AI caps.

Column	Type	Key	Null	Note
setting_key	VARCHAR(60)	PK		
value	JSON			
updated_by	BIGINT UNSIGNED	FK	yes	
updated_at	DATETIME(3)			